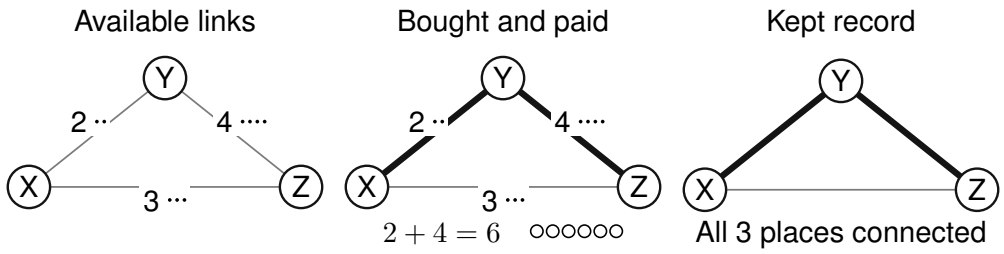
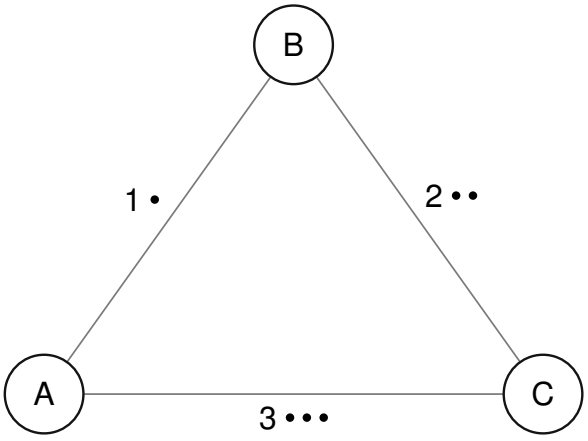
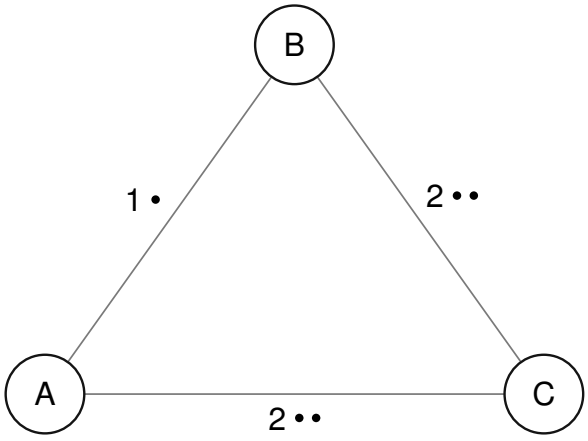
Mark whole printed links you buy. Thick links are bought. Pay each link's price once. A price of 2, shown as ••, costs two counters. Return a link to get its counters back. Travel follows bought links and turns only at circled places.



Problem 1: Connect all three places for the fewest counters on each map.

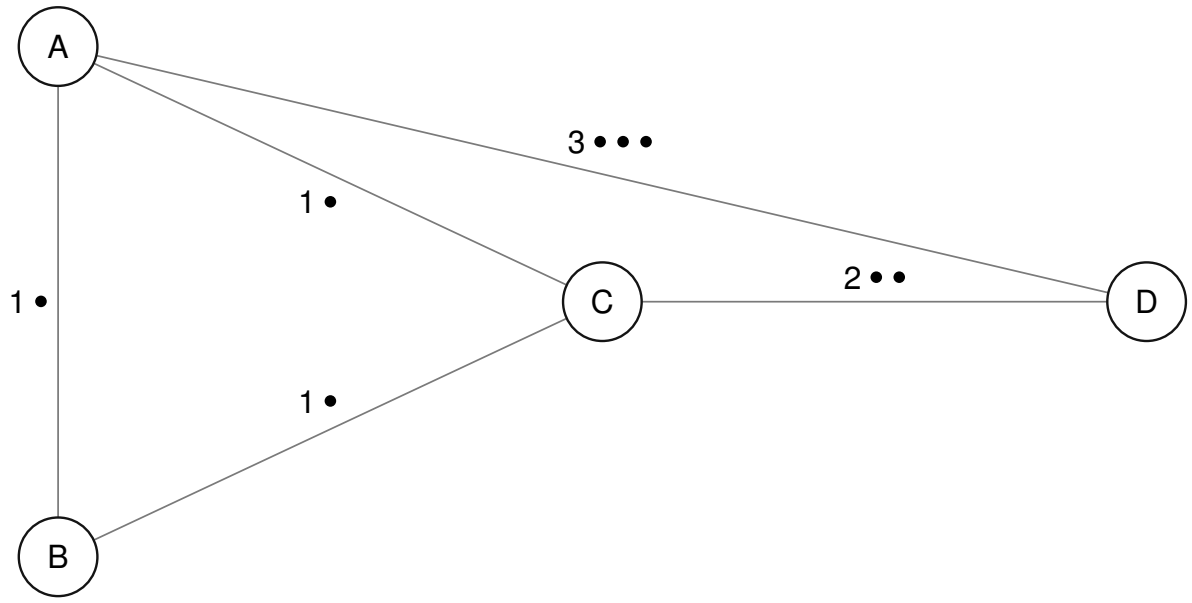


Cost: _____

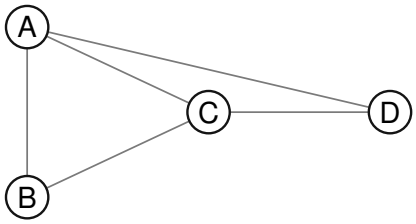


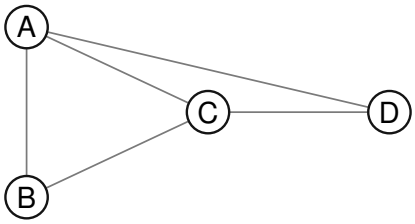
Cost: _____

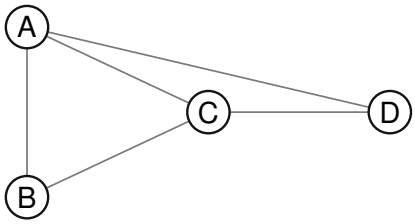
Problem 2: Find every cheapest purchase that connects all four places.

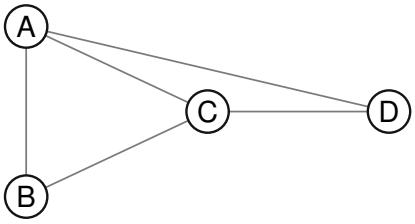


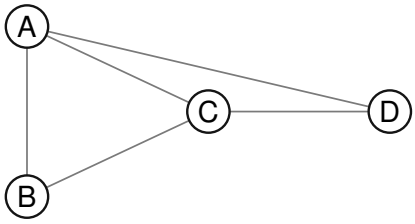
Cost: _____



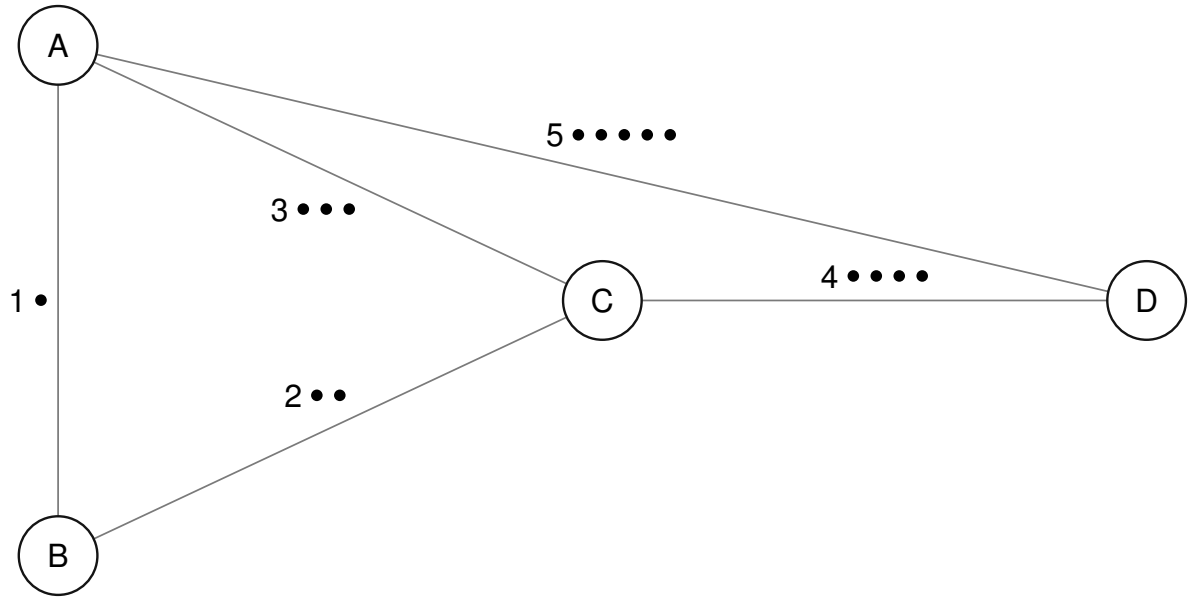




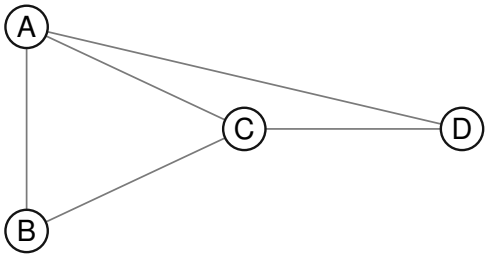


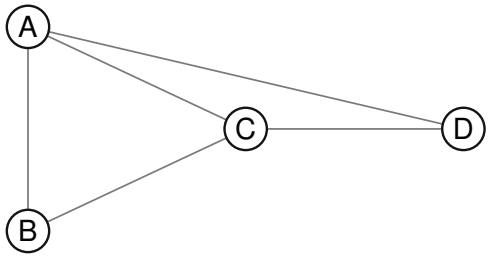


Problem 3: Connect all four places for the least total cost. Show why a cheaper purchase cannot work.

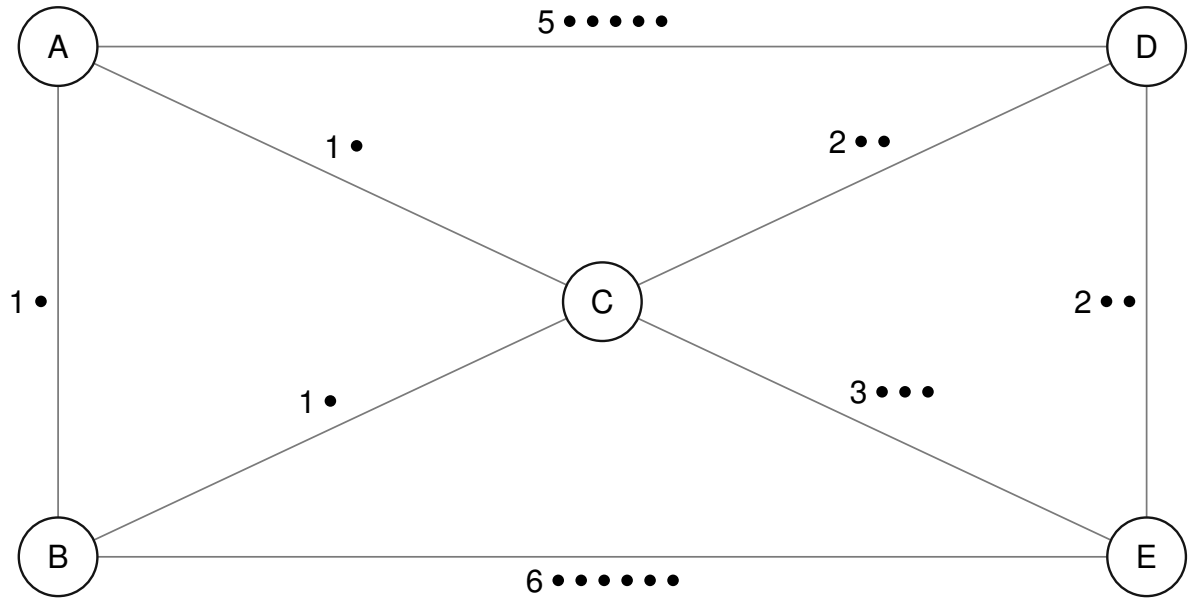


Cost: _____

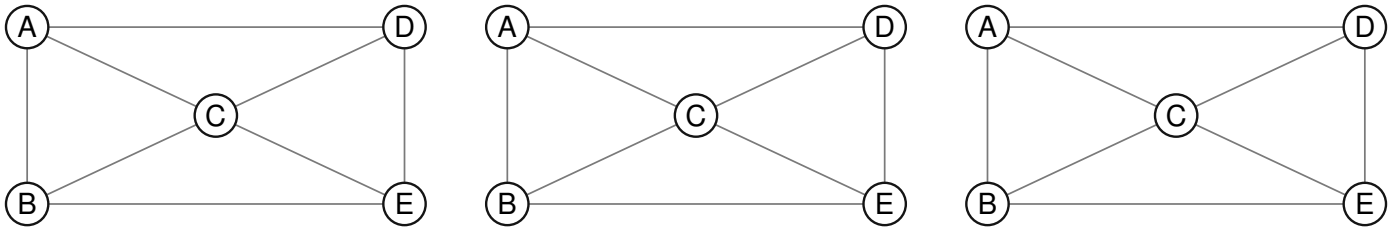




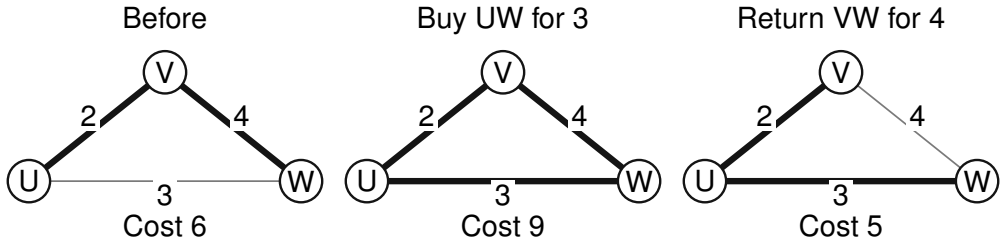
Problem 4: Find every cheapest purchase that connects all five places.

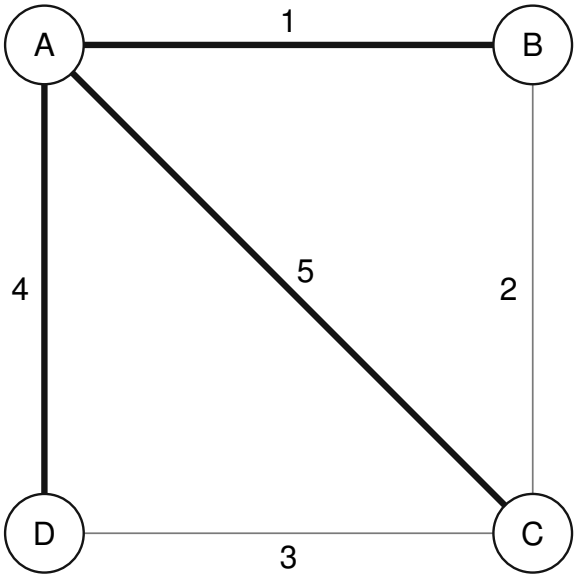


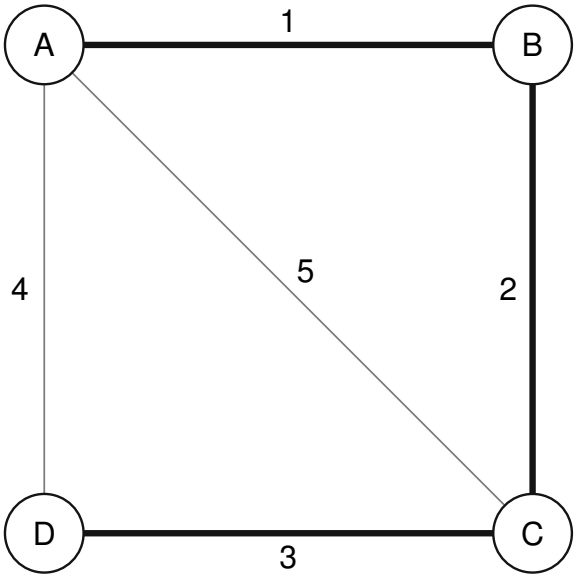
Cost: _____



Problem 5: Starting each time from the printed thick purchase, find every cheaper one-link swap: buy one unused link and return one bought link. The result must connect every place.



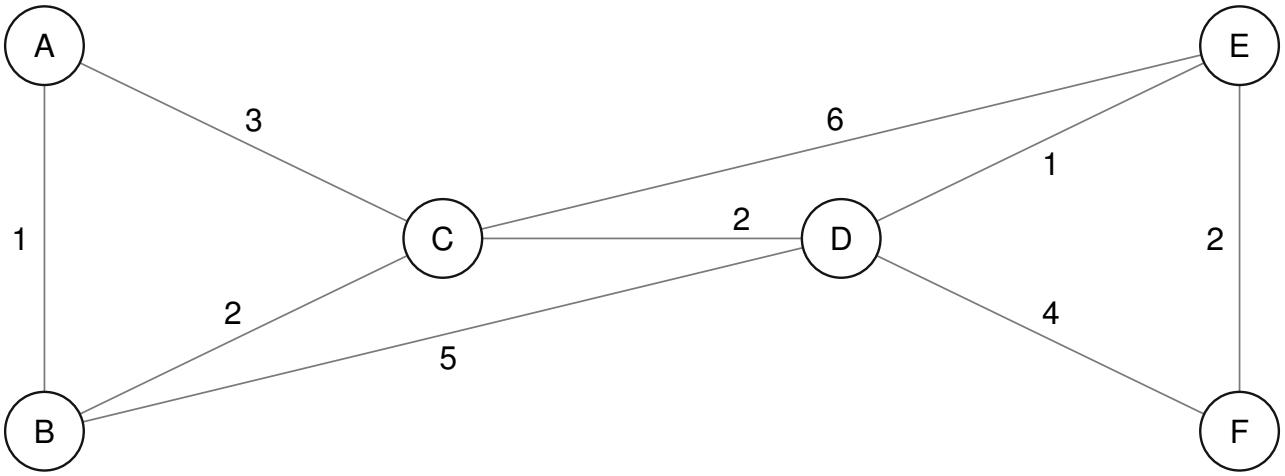




Return	Buy	New cost

Return	Buy	New cost

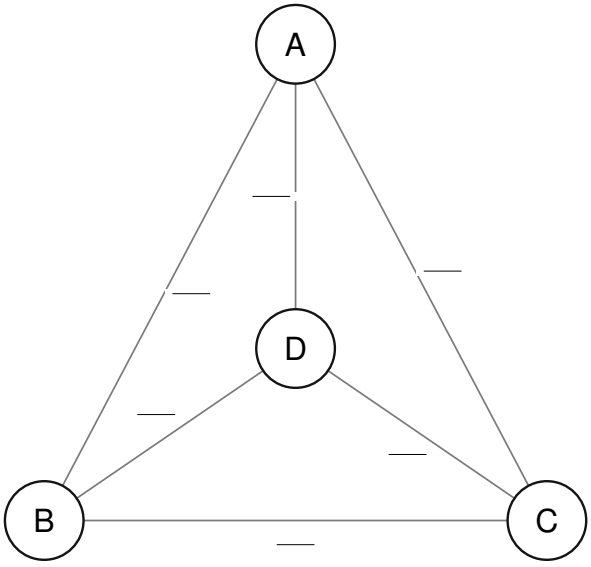
Problem 6: Which links must be in every cheapest purchase? Which links are in no cheapest purchase? Give reasons that rule out every competing purchase.

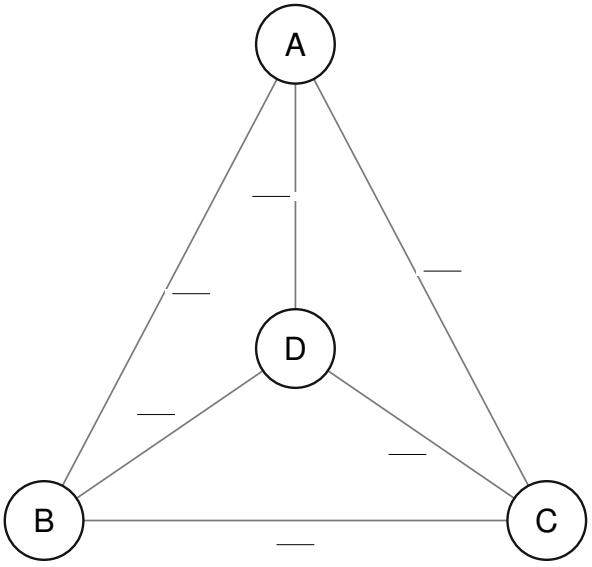


In every cheapest purchase:

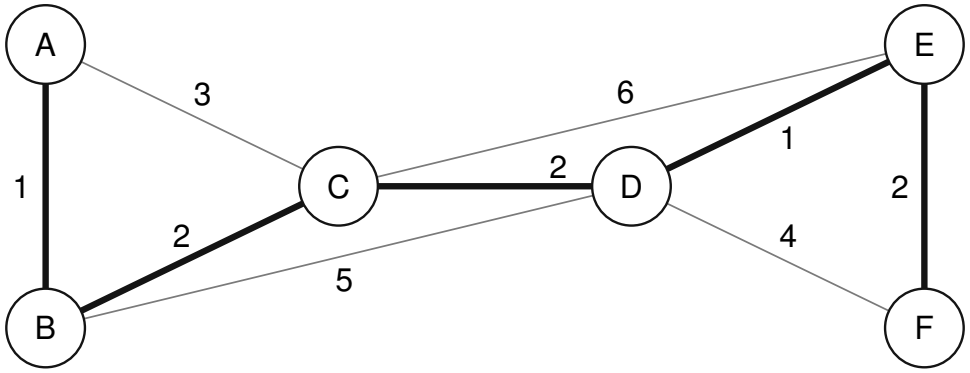
In no cheapest purchase:

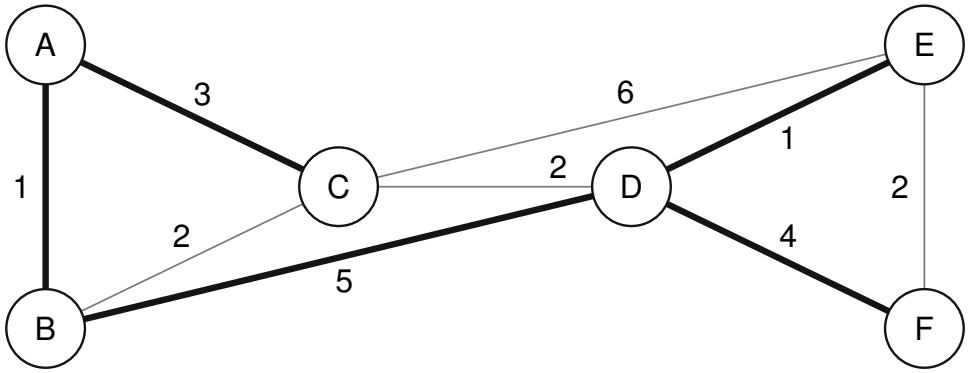
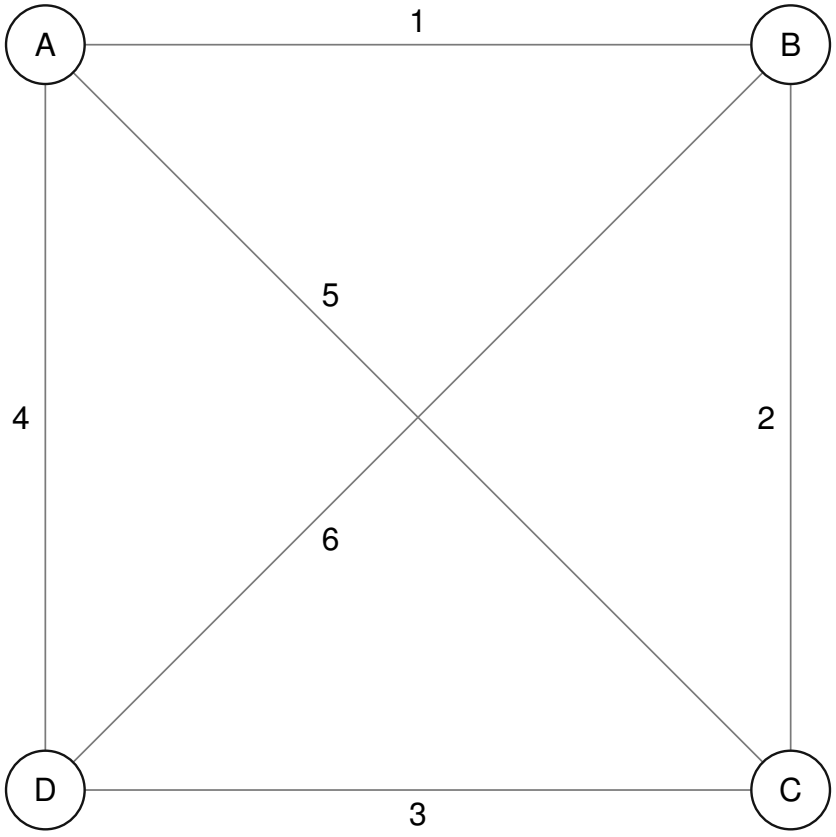
Problem 7: Give each link a price of 1, 2, 3 or 4. Make the left map have exactly one cheapest purchase and the right map have at least two. Show why both meet their condition.





Problem 8: All link prices are positive. A loop follows bought links back to its start without using a link twice. Can a cheapest purchase contain a loop? For each thick purchase, find a cheaper one-link swap or explain why none exists. Can a connected purchase with no loop cost more than the cheapest one even though no one-link swap makes it cheaper?





Problem 9: Find the cheapest purchase on this map. Can any map with a different positive price on every link have two different cheapest purchases? Explain your answer.